School of Natural Sciences and Mathematics

The Biology Program at UT Dallas emphasizes the unifying molecular and cellular nature of organisms. At the center of the Biology undergraduate curriculum are the biochemical, genetic, and cell biology concepts and tools used to study the genes of prokaryotes and eukaryotes, to study the proteins and ribonucleic acids (RNA) encoded by these genes, and to study how the expression of these genes is regulated during the development and lifetimes of organisms. Molecular Biology represents a fusion of the four disciplines of biochemistry, biophysics, genetics, and cell biology. Modern biology requires a background in other disciplines such as chemistry, mathematics, physics, and computer sciences. Principles from these disciplines have to be merged to understand and apply new biotechnology and genetic engineering techniques. It is desirable for entering students to have a broad interest and background in the sciences.

Biology (BA, BS)

Both BS and BA degrees are offered in Biology at UT Dallas. The BS degrees are intended as preparation for scientific careers in biology or careers in the health professions. The BA degree is intended as a liberal arts biology major with less emphasis on calculus and more semester credit hours free for coursework in other disciplines. Biology offers a streamlined double major with Business Administration or Criminology. Accelerated BS/MS Fast Track programs are available that provide the opportunity to partially satisfy the requirements of a master's degree while completing the bachelor's degree in Biology.

Bachelor of Arts in Biology

Degree Requirements (120 semester credit hours)

View an Example of Degree Requirements by Semester

Faculty

FACG> nsm-biology-ba

Professors: Juan E. González, Tae Hoon Kim, Kelli Palmer, Lawrence J. Reitzer, Donal Skinner, Stephen Spiro, Li Zhang, Michael Qiwei Zhang

Associate Professors: Joseph Boll, Nicole De Nisco, Nikki Delk, Tian Hong, Faruck Morcos, Duane D. Winkler, Zhenyu Xuan

Assistant Professors: Nicholas Dillon, Xintong Dong, Lin Jia, Purna Joshi, Brandon Kim, Erica Sanchez, Darshan Sapkota, Yuki Shindo, Jon Sin

Professors Emeriti: Hans Bremer, Lee A. Bulla, Rockford Draper, Donald M. Gray

Associate Professors Emeriti: Gail A. M. Breen, Dennis L. Miller

Clinical Professor: David Murchison, Eberhard Voit

Professors of Instruction: Mehmet Candas, Elizabeth Pickett, Uma Srikanth

Associate Professors of Instruction: Meenakshi Maitra, Jing Pan, Eva Sadat, Subha Sarcar, Michelle Wilson, Zhuoru Wu, Wen-Ho Yu

Assistant Professors of Instruction: Stephanie Boyd, Andy Cheshire, Anne Davenport, Matthew Esparza, Amy Jo Gomez, Yi Huang, Li Liu, Simbarashe Mazambani, Iti Mehta, Ritu Mishra, Ramesh Padmanabhan, Narges Salamat

Research Associate Professor: Ru-Hung Wang

Lecturer: Kathleen McRoy

I. Core Curriculum Requirements: 42 semester credit hours

Curriculum Requirements can be fulfilled by other approved courses from institutions of higher education. The courses listed are recommended as the most efficient way to satisfy both Core Curriculum and Major Requirements at UT Dallas.

Communication: 6 semester credit hours

Select any 6 semester credit hours from Communication Core courses (see advisor)

Mathematics: 3 semester credit hours

Choose one course from the following:

MATH 2413 Differential Calculus[1, 2, 3]

or MATH 2417 Calculus I[1, 2, 3]

MATH 1325 Applied Calculus I[1] (BA only)

Or select any 3 semester credit hours from Mathematics Core courses (see advisor)

Life and Physical Sciences: 6 semester credit hours

CHEM 1311 General Chemistry I[1]

or CHEM 1315 Honors Freshman Chemistry I[1]

CHEM 1312 General Chemistry II[1]

or CHEM 1316 Honors Freshman Chemistry II[1]

Or select any 6 semester credit hours from Life and Physical Sciences Core courses (see advisor)

Language, Philosophy and Culture: 3 semester credit hours

Select any 3 semester credit hours from Language, Philosophy and Culture Core courses (see advisor)

Creative Arts: 3 semester credit hours

Select any 3 semester credit hours Creative Arts Core courses (see advisor)

American History: 6 semester credit hours

Select any 6 semester credit hours from American History Core courses (see advisor)

Government/Political Science: 6 semester credit hours

Select any 6 semester credit hours from Government/Political Science Core courses (see advisor)

Social and Behavioral Sciences: 3 semester credit hours

Select any 3 semester credit hours from Social and Behavioral Sciences Core courses (see advisor)

Component Area Option: 6 semester credit hours

Choose two courses from the following:

BIOL 2311 Introduction to Modern Biology I[1, 4]

MATH 2414 Integral Calculus[1, 2, 3]

or MATH 2419 Calculus II[1, 2, 3]

or STAT 2332 Introductory Statistics for Life Sciences[1]

Or select any 6 semester credit hours from Component Area Option Core courses (see advisor)

II. Major Requirements: 54-57 semester credit hours beyond Core Curriculum

Major Preparatory Courses: 19-22 semester credit hours beyond Core Curriculum

NATS 1101 Natural Sciences and Mathematics Freshman Seminar

CHEM 1311 General Chemistry I[1]

or CHEM 1315 Honors Freshman Chemistry I[1]

CHEM 1111 General Chemistry Laboratory I

or CHEM 1115 Honors Freshman Chemistry Laboratory I

CHEM 1312 General Chemistry II[1]

or CHEM 1316 Honors Freshman Chemistry II[1]

CHEM 1112 General Chemistry Laboratory II

or CHEM 1116 Honors Freshman Chemistry Laboratory II

CHEM 2323 Introductory Organic Chemistry I[4]

or CHEM 2327 Honors Organic Chemistry I[4]

CHEM 2325 Introductory Organic Chemistry II[4]

or CHEM 2328 Honors Organic Chemistry II[4]

CHEM 2233 Introductory Organic Chemistry Laboratory[4]

or CHEM 2237 Honors Organic Chemistry Laboratory[4]

MATH 2413 Differential Calculus[5, 1, 2, 3]

and MATH 2414 Integral Calculus[5, 1, 2, 3]

or MATH 2417 Calculus I[5, 1, 2, 3]

and MATH 2419 Calculus II[5, 1, 2, 3]

or MATH 1325 Applied Calculus I[5, 1, 2, 3] (BA only)

and STAT 2332 Introductory Statistics for Life Sciences[1, 2]

PHYS 2325 Mechanics[5, 1]

and PHYS 2125 Physics Laboratory I[5, 1] or PHYS 2421 Honors Physics I - Mechanics and Heat[6]

or PHYS 1301 College Physics I

and PHYS 1101 College Physics Laboratory I

PHYS 2326 Electromagnetism and Waves

and PHYS 2126 Physics Laboratory II

or PHYS 2422 Honors Physics II - Electromagnetism and Waves

and PHYS 2126 Physics Laboratory II

or PHYS 1302 College Physics II

and PHYS 1102 College Physics Laboratory II

Major Core Courses: 26 semester credit hours beyond Core Curriculum

BIOL 2281 Introductory Biology Laboratory[4]

or BIOL 2V81 Course-based Undergraduate Research Experience in Biology

BIOL 2111 Introduction to Modern Biology Workshop I[4]

BIOL 2112 Introduction to Modern Biology Workshop II[4]

BIOL 2311 Introduction to Modern Biology I[1, 4]

BIOL 2312 Introduction to Modern Biology II[4]

BIOL 3401 Genetics

BIOL 3402 Molecular and Cell Biology

BIOL 3461 Biochemistry I

or CHEM 3461 Biochemistry I

BIOL 3462 Biochemistry II

or CHEM 3462 Biochemistry II

or BIOL 3335 Microbial Physiology

BIOL 3380 Biochemistry Laboratory

Major Related Courses: 9 semester credit hours[7]

9 semester credit hours of upper-division BIOL electives

III. Elective Requirements: 18-24 semester credit hours

The plan must include sufficient upper-division courses to total 45 upper-division semester credit hours.

Bachelor of Science in Biology

Degree Requirements (120 semester credit hours)

View an Example of Degree Requirements by Semester

Faculty

FACG> nsm-biology-bs

Professors: Juan E. González, Tae Hoon Kim, Kelli Palmer, Lawrence J. Reitzer, Donal Skinner, Stephen Spiro, Li Zhang, Michael Qiwei Zhang

Associate Professors: Joseph Boll, Nicole De Nisco, Nikki Delk, Tian Hong, Faruck Morcos, Duane D. Winkler, Zhenyu Xuan

Assistant Professors: Nicholas Dillon, Xintong Dong, Lin Jia, Purna Joshi, Brandon Kim, Erica Sanchez, Darshan Sapkota, Yuki Shindo, Jon Sin

Professors Emeriti: Hans Bremer, Lee A. Bulla, Rockford Draper, Donald M. Gray

Associate Professors Emeriti: Gail A. M. Breen, Dennis L. Miller

Clinical Professor: David Murchison, Eberhard Voit

Professors of Instruction: Mehmet Candas, Elizabeth Pickett, Uma Srikanth

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Assistant Professors of Instruction: Stephanie Boyd, Andy Cheshire, Anne Davenport, Matthew Esparza, Amy Jo Gomez, Yi Huang, Li Liu, Simbarashe Mazambani, Iti Mehta, Ritu Mishra, Ramesh Padmanabhan, Narges Salamat

Research Associate Professor: Ru-Hung Wang

Lecturer: Kathleen McRoy

I. Core Curriculum Requirements: 42 semester credit hours

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Select any 6 semester credit hours from Communication Core courses (see advisor)

Mathematics: 3 semester credit hours

Choose one course from the following:

MATH 2413 Differential Calculus[1, 2, 3]

or MATH 2417 Calculus I[1, 2, 3]

Or select any 3 semester credit hours from Mathematics Core courses (see advisor)

Life and Physical Sciences: 6 semester credit hours

CHEM 1311 General Chemistry I[1]

or CHEM 1315 Honors Freshman Chemistry I[1]

CHEM 1312 General Chemistry II[1]

or CHEM 1316 Honors Freshman Chemistry II[1]

Or select any 6 semester credit hours from Life and Physical Sciences Core courses (see advisor)

Language, Philosophy and Culture: 3 semester credit hours

Select any 3 semester credit hours from Language, Philosophy and Culture Core courses (see advisor)

Creative Arts: 3 semester credit hours

Select any 3 semester credit hours Creative Arts Core courses (see advisor)

American History: 6 semester credit hours

Select any 6 semester credit hours from American History Core courses (see advisor)

Government/Political Science: 6 semester credit hours

Select any 6 semester credit hours from Government/Political Science Core courses (see advisor)

Social and Behavioral Sciences: 3 semester credit hours

Select any 3 semester credit hours from Social and Behavioral Sciences Core courses (see advisor)

Component Area Option: 6 semester credit hours

Choose two courses from the following:

BIOL 2311 Introduction to Modern Biology I[1, 4]

MATH 2414 Integral Calculus[1, 2, 3]

or MATH 2419 Calculus II[1, 2, 3]

Or select any 6 semester credit hours from Component Area Option Core courses (see advisor)

II. Major Requirements: 62-63 semester credit hours beyond Core Curriculum

Major Preparatory Courses: 21-22 semester credit hours beyond Core Curriculum

NATS 1101 Natural Sciences and Mathematics Freshman Seminar

CHEM 1311 General Chemistry I[1]

or CHEM 1315 Honors Freshman Chemistry I[1]

CHEM 1111 General Chemistry Laboratory I

or CHEM 1115 Honors Freshman Chemistry Laboratory I

CHEM 1312 General Chemistry II[1]

or CHEM 1316 Honors Freshman Chemistry II[1]

CHEM 1112 General Chemistry Laboratory II

or CHEM 1116 Honors Freshman Chemistry Laboratory II

CHEM 2323 Introductory Organic Chemistry I[4]

or CHEM 2327 Honors Organic Chemistry I[4]

CHEM 2325 Introductory Organic Chemistry II[4]

or CHEM 2328 Honors Organic Chemistry II[4]

CHEM 2233 Introductory Organic Chemistry Laboratory[4]

or CHEM 2237 Honors Organic Chemistry Laboratory[4]

MATH 2413 Differential Calculus[5, 1, 2, 3]

and MATH 2414 Integral Calculus[5, 1, 2, 3]

or MATH 2417 Calculus I[5, 1, 2, 3]

and MATH 2419 Calculus II[5, 1, 2, 3]

PHYS 2325 Mechanics[5, 1]

and PHYS 2125 Physics Laboratory I[5, 1]

or PHYS 2421 Honors Physics I - Mechanics and Heat[6]

or PHYS 1301 College Physics I

and PHYS 1101 College Physics Laboratory I

PHYS 2326 Electromagnetism and Waves

and PHYS 2126 Physics Laboratory II

or PHYS 2422 Honors Physics II - Electromagnetism and Waves

and PHYS 2126 Physics Laboratory II

or PHYS 1302 College Physics II

and PHYS 1102 College Physics Laboratory II

Major Core Courses: 29 semester credit hours beyond Core Curriculum

BIOL 2281 Introductory Biology Laboratory[4]

or BIOL 2V81 Course-based Undergraduate Research Experience in Biology

BIOL 2111 Introduction to Modern Biology Workshop I[4]

BIOL 2112 Introduction to Modern Biology Workshop II[4]

BIOL 2311 Introduction to Modern Biology I[1, 4]

BIOL 2312 Introduction to Modern Biology II[4]

BIOL 3401 Genetics

BIOL 3402 Molecular and Cell Biology

BIOL 3461 Biochemistry I

or CHEM 3461 Biochemistry I

BIOL 3462 Biochemistry II

or CHEM 3462 Biochemistry II

or BIOL 3335 Microbial Physiology

BIOL 3380 Biochemistry Laboratory

BIOL 4380 Cell and Molecular Biology Laboratory

Major Related Courses: 12 semester credit hours[7]

12 semester credit hours of upper-division BIOL electives

III. Elective Requirements: 15-16 semester credit hours

The plan must include sufficient upper-division courses to total 45 upper-division semester credit hours.

UTeach Option

The UTeach option may be added to either the BA or BS degree. UTeach Dallas Option degree plans are streamlined to allow students to complete both a rigorous Bachelor of Science or Bachelor of Arts degree and all coursework for middle or high school teacher certification in four years. Teaching Option degrees require deep content knowledge combined with courses grounded in the latest research on math and science education. While most graduates go on to classroom teaching, UTeach alums are also prepared to enter graduate school and to work in a discipline related industry.

Fast Track Baccalaureate/Master's Degrees

For students with strong academic record who intend to pursue master's studies at UT Dallas, accelerated BS/MS Fast Tracks are available. After Fast Track admission to an MS program, students may take up to 15 semester credit hours of approved graduate courses in their senior year to use toward completion of both the BS and MS degrees. The available Fast Track options and their admission requirements, if any, in addition to having completed 90 or more semester credit hours (out of which 36 hours are from the core curriculum) with a cumulative GPA of at least 3.00, are as follows:

- MS in Molecular and Cell Biology: Students must have a GPA of at least 3.20 and have completed BIOL 2311, BIOL 2312, and BIOL 2281, or equivalent courses.
- MS in Biotechnology: Students must have a GPA of at least 3.20 and have completed BIOL 2311, BIOL 2312, and BIOL 2281, or equivalent courses.
- MS in Bioinformatics and Computational Biology: Students must have a cumulative GPA of at least 3.20 in all mathematics and statistics courses.
- MS in Data Science and Statistics: (Only Data Science and Applied Statistics specializations) Students must have a cumulative GPA of at least 3.20 in all mathematics and statistics courses.

Interested students, after reviewing the UT Dallas Fast Track policy, should contact their undergraduate advisor and the graduate advisor of the intended MS program well in advance of their junior year to prepare a course sequence permitting maximal advantage and apply to the Fast Track program.

Degree Planning

Upper-division biology courses taken at other institutions may be included as part of the degree plan subject to the provisions of the section on Transfer Admissions.

Major-related courses may not include more than 9 semester credit hours (BS) or 6 semester credit hours (BA) of upper-division transfer credit and not more than 3 semester credit hours (Biology major) or 6 semester credit hours (Molecular Biology major) of individual instruction (e.g., BIOL 3V90, BIOL 3V91, BIOL 3V96, BIOL 4302, BIOL 4390, BIOL 4391, BIOL 4399, or BIOL 4V99).

Students planning a career in a particular allied health profession should consult the school they expect to attend to apprise themselves of the course requirements for admission.

Admission standards for medical and dental schools are set by the individual professional school, whose specific requirements should be reviewed with the help of the UT Dallas Health Professions Advising Center (HPAC). Most professional schools prefer that admission applications be channeled through the HPAC.

Minors

Students must take a minimum of 18 semester credit hours for the minor, 12 of which must be upper-division semester credit hours. Students who take a minor will be expected to meet the normal prerequisites in courses making up the minor, and should maintain a minimum GPA of 2.000 on a 4.00 scale (C average). Semester credit hours may not be used to satisfy both the major and minor requirements; however, free elective semester credit hours or major preparatory classes may be used to satisfy the minor.

For all minors in the School of Natural Sciences and Mathematics students must complete all prerequisite sequences for required minor courses.

Minor in Biology

18 semester credit hours

Required: 12 semester credit hours

BIOL 2311 Introduction to Modern Biology I

BIOL 2111 Introduction to Modern Biology Workshop I

BIOL 3401 Genetics

BIOL 3461 Biochemistry I

or CHEM 3461 Biochemistry I

Also:

Two BIOL approved electives for majors

Minor in Biomolecular Structure

18 semester credit hours

Required: 13 semester credit hours

BIOL 3336 Protein and Nucleic Acid Structure

or BIOL 4305 Molecular Evolution

BIOL 4461 Biophysical Chemistry (unless taken to fulfill the Molecular Biology major requirements)

CHEM 2323 Introductory Organic Chemistry I

CHEM 2325 Introductory Organic Chemistry II

Also:

A minimum of 5 additional semester credit hours in approved BIOL, CHEM, CS, EE, MATH, or PHYS electives

Minor in Microbiology

18 semester credit hours

Required: 15 semester credit hours

BIOL 3303 Introduction to Microbiology[8]

and BIOL 3203 Introduction to Microbiology Lab

BIOL 3335 Microbial Physiology[9]

BIOL 4350 Medical Microbiology

BIOL 4345 Immunobiology

CHEM 2323 Introductory Organic Chemistry I

Also:

One approved microbiology elective

Minor in Molecular and Cell Biology

18 semester credit hours

Required: 6 semester credit hours

CHEM 2323 Introductory Organic Chemistry I

CHEM 2325 Introductory Organic Chemistry II

Also:

Four approved molecular and cell biology electives

Minor in Neurobiology

18 semester credit hours

Required: 15 semester credit hours

CHEM 2323 Introductory Organic Chemistry I

CHEM 2325 Introductory Organic Chemistry II

BIOL 4356 Molecular Neuropathology

or BIOL 4357 Molecular Neuropathology II

NSC 4353 Neuroscience Laboratory Methods

NSC 4354 Integrative Neuroscience

or NSC 4352 Cellular Neuroscience

Also:

One BIOL approved electives for majors, must be BIOL (not NSC or CHEM)

1. A required Major course that also fulfills a Core Curriculum requirement. Semester credit hours are counted in Core Curriculum.

2. Six semester credit hours of Calculus are counted under Mathematics Core and Component Area Option and 2 semester credit hours of Calculus are counted as Major Preparatory Courses.

3. Students may substitute MATH 2413 and MATH 2414 by taking MATH 2417 and MATH 2419.

4. Indicates a prerequisite class to be completed before enrolling for upper-division classes.

5. Curriculum Requirements can be fulfilled by other approved courses from institutions of higher education. The courses listed are recommended as the most efficient way to satisfy both Core Curriculum and Major Requirements at UT Dallas.

6. Students who complete PHYS 2421 do not need to complete PHYS 2125

7. Up to 3 semester credit hours of individual instruction may be used in fulfilling this requirement for BA degree. Up to 6 semester credit hours of individual instruction may be used in fulfilling this requirement for BS degree.

8. Two semester credit hours of BIOL 3303 may be used to satisfy the upper-division elective requirement for Biology and Molecular Biology majors.

9. May be substituted with CHEM 2325 if used to satisfy the Biochemistry II core requirement for

Biology and Molecular Biology majors.

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